Participants for all studies will be recruited from the psychology research participation pools at UQ and UWA. Prior to running each study, we will conduct a large-scale parameter recovery analysis to assess model identifiability, and to determine the number of participants required for reliable parameter estimation using BayesFlow. Based on experience with similar experiments in our lab, we estimate that we will need 100 participants per study.

Study 1: Prioritisation

Experiment. The goal of Study 1 is to develop a basic model of workload management in an environment where people manage multiple tasks with varying deadlines. Data for modeling will come from a modified version of the block management task developed by Tulga and Sheridan [13, 14]. Blocks appear on the screen and move to the right along a conveyor belt (see Figure 1). Each block represents a task that needs to be completed. The task is to eliminate the block before it reaches the end of the conveyor belt (the deadline). Participants are able to select a block by clicking on it with their mouse. When selected, the right edge of the block is eaten away at a fixed rate. The width of the block, therefore, indicates the time required to complete the task. The colour of the block indicates the number of points earned by eliminating the block (task value). Time required, deadline and value are allowed to vary (see below). Participants need to periodically reassess the priority of the different blocks, because new blocks can appear that have an earlier deadline or greater value than the existing blocks.

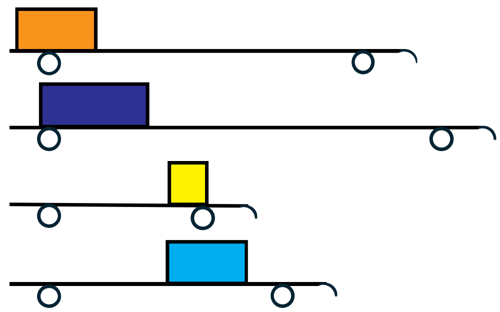

Figure 1: Block management task.

Time pressure will be manipulated within participants by varying the number of blocks on the screen, together with the time required and time available for each block and the overlap in required activity between blocks. Time pressure will follow a cyclic pattern, with the phase of the cycle counterbalanced across participants. For each participant, we have a time series recording which block they were working on, and when they switched. Workload will be measured using the Detection Response Task [15], which is an ISO standard secondary-task measure of workload, and self-report (on a Likert scale from 1-10 see [16]). Self-reported workload will be collected at 2 minute intervals.

Model development. Study 1 will allow us to develop the basics of a broad model that can account for the order in which participants do tasks, the frequency and timing of task switches, and workload (as measured by both the DRT and self-report). We treat prioritisation as a continuous evidence accumulation process, in which participants have an evolving representation of the priority of each task. Figure 2 shows an example involving two task attributes: time to deadline and value. These attributes are translated into a set of priorities via a set of weights that may vary across people or over time. In this example, block A has the highest initial priority, because it is a high value block that is close to the deadline. For this reason, the person starts working on block A, however, the priority for block D overtakes that for block A before A is completed, so the person switches. The panel on the right shows the estimated workload for alternative models of workload (i.e, whether workload is determined solely by the focal task, or globally from all outstanding tasks).

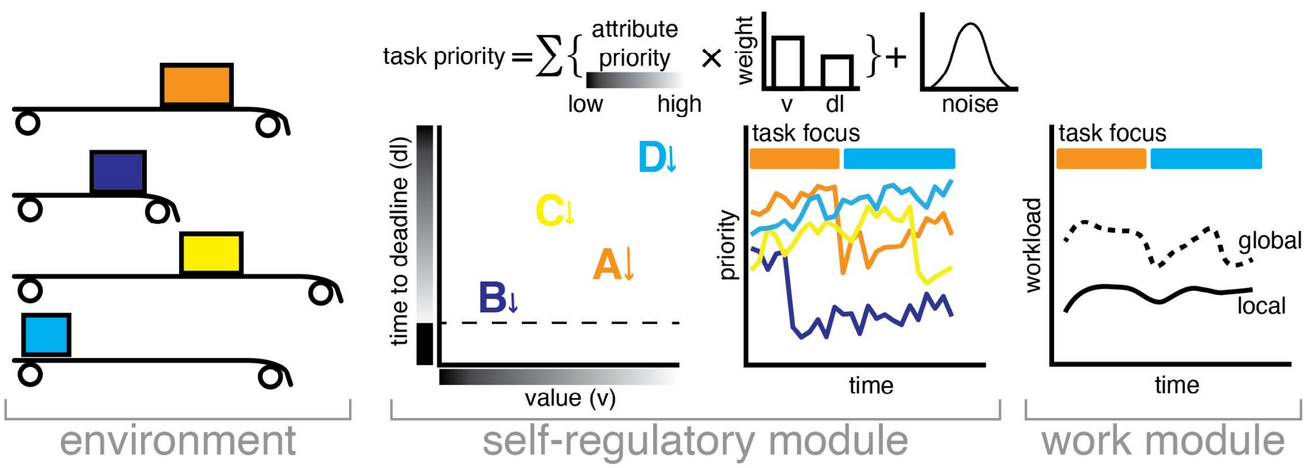

Figure 2: Schematic representation of proposed model for Study 1.

Whilst the block management task is relatively simple, there are many potential models of this task that differ in their assumptions. Model development will focus on using the data to competitively test different combinations of assumptions. One major question is how responsive priority updates are to changes in the environment. While task priorities may evolve continuously in response to approaching deadlines, it is likely that accurately updating information on the tasks will require some mental effort that will take time away from processing blocks or making switch decisions, and how that happens will be explored in different variants of the model that treat this as a time cost versus a sharing of resources. A second question is the role for scheduling in this task. As described, and following precedence in models of goal pursuit [10], people may simply evaluate tasks independently. However, people may adapt more advanced scheduling strategies [14] and this strategic aspect---which will also trade off against time or resources available for other tasks---will be addressed in model developments. For example, scheduling may be enacted in this task by controlling the weights on attributes to focus (for example) on the task next closest to completion. Completing a block may trigger a reassessment of plans, and then updating of weights in light of the current state. A third question is how workload itself is calculated; as shown in Figure 3, workload might be calculated just from the focal task ("local"), or on the basis of all outstanding tasks ("global"), and different measures of workload (DRT vs self-report) may be sensitive to different metrics, indicating that they tap into different processes. Furthermore, workload may itself impact on scheduling and task completion; for example, high workload may shift people into a more basic style of processing where weight is fully given to only one or attributes, such as focussing only on time to deadline to prioritise tasks [7]. Questions such as these cannot be answered solely on the basis of data, and highlight the importance and necessity of model-based analysis.

Studies 2-4: Effort & Strategy

Experiments. Studies 2 and 3 will extend the basic model to explain how people prioritise the allocation of their time when managing tasks that require effort. The block management task will be modified in Study 2 so that physical effort is required to eliminate the blocks. Participants will be repeatedly click on the block to eliminate it, each click removing a certain amount of the block. The speed of clicking provides a behavioural measure of effort [17]. We will use the same manipulation of time pressure as in Study 1. Motivation will be manipulated using a combination of instructions ("try your hardest" vs "take it easy") and incentives.

In Study 3, the block management task will be modified so that it requires cognitive rather than physical effort. Rather than making clicks, participants will be required to complete basic cognitive tasks that have been well validated in the effort discounting literature, such as choice response time tasks (e.g., random dot kinematograms), or solving anagrams [18]. Participants will be required to complete a specified number of operations to complete each block. The difficulty of the blocks will be varied, by changing the task itself or its difficulty.

Study 4 will expand on Study 3 by allowing participants to adjust their strategy. This will be accomplished by implementing a nonlinear value function so that there are diminishing returns from continuing to work on a task. This will allow us to examine the relationship between workload and strategy. For example, an optimal strategy may be to half-complete a number of tasks, but regulation of workload may lead participants to effectively focus on only a few tasks and complete those to a high level. Study 4 will also provide critical information on how workload is calculated; although there are deadlines, blocks cannot technically be completed, and so there is no sense in which a deadline can be missed.

Model development. Studies 2-4 will allow us to refine the model, adding additional mechanisms to account for the different ways that people are able to regulate workload in complex environments. We do this by treating the work itself (clicking on the blocks or completing basic cognitive tasks) as an accumulation process, with three parameters: a rate, threshold, and delay. The rate represents the speed of processing, which is an indicator of effort. The threshold represents the amount of work that needs to be done, while the delay represents the amount of time that the person requires to orient themselves to the task before commencing work (the cost of switching from one task to another). As indicated earlier, we assume that effort increases in response to time pressure, until the person reaches the maximum amount of effort that they are prepared to apply to the task. The point of maximum motivational potential should be higher in the high motivation condition than in the low motivation condition, and should gradually decline over the course of the experiment as participants become more fatigued. In Study 4, we extend the model by allowing the threshold to vary in order to capture strategy, and examine whether there are individual differences in how people regulate workload (i.e., whether some people respond by changing how they prioritise, others respond by increasing effort, and others by changing strategy).